



NORTH SOUTH UNIVERSITY

Centre of Excellence in Higher Education

DEPARTMENT OF MATHEMATICS AND PHYSICS

School of Engineering and Physical Sciences

Course outline: Spring 2025

Course Title	Calculus and Analytical Geometry III (3 Credits)
Course Code	MAT 250
Section:	

Instructor Name	Hasina Akter
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Office Room	SAC 1021
Office Hours	ST 2:00pm – 4:00pm, MW 2:00pm – 4:00pm, R 2:00pm – 4:00pm and by appointment

Prerequisites	MAT 130
Course Objectives	1. To demonstrate the function of several variables and plotting 3D figures. 2. To teach the concept of partial derivatives and their applications. 3. To develop the ability of multiple integrations in different coordinate systems. 4. To analyze the vector calculus and its physical significance.
Course Learning Outcomes	Upon the successful completion of this course, a student will be able to: (CO-1) Classify the difference between single and several variables functions and limits as well as plotting 3D figures. (CO-2) Evaluate the partial derivatives for several variables functions and distinguish ordinary and partial derivatives. (CO-3) Apply multiple integration techniques to find the area and volume of the different model geometries. (CO-4) Demonstrate their understanding of vector calculus and vector algebra. (CO-5) Apply line and surface integrals to evaluate the work done and the corresponding flux.

Text Book	
Author	Howard Anton, IRL Bivens, and Stephen Davis
Title	Calculus: Early Transcendentals
Edition & Year	10 th Edition, 2013
Publisher	John Wiley & Sons, Inc

Mapping of Course Outcomes

#	Course Outcomes (CO)	Bloom's taxonomy domain/level (C: Cognitive P: Psychomotor A: Affective)	Delivery methods and activities	Assessment tools
CO-1	Classify the difference between single and several variables functions and limits as well as plotting 3D figures.	C1, C2, C3	Lecture Discussion	Quiz, Assignment
CO-2	Evaluate the partial derivatives for several variables functions and distinguish ordinary and partial derivatives.	C3, C4, P2	Lecture, in-class group discussion,	Concept clarification, Midterm exam, Assignment
CO-3	Apply multiple integration techniques to find area and volume of the different model geometries.	C2, C3, P2	Lecture, Discussion	Class work, Quiz, Assignment, Final Exam
CO-4	Demonstrate their understanding of vector calculus and vector algebra.	C2, P2	Lecture, Discussion	Concept, Demonstration, Quiz, Assignment, Final Exam
CO-5	Apply line and surface integrals to evaluate the work done and the corresponding flux.	C2, C3, C4, P2	Lecture Demonstration	Assignment, Final Exam

Assessment Strategy		Grading Policy*		
Grading tool	Percentages (%)	Numerical Scores	Letter Grade	Grade points
Attendance	05%	93 and above	A Excellent	4.0
Assignments (Average of 2) *	10%	90 - 92	A-	3.7
Quizzes (Best 2 out of 4)**	20%	83 - 86	B Good	3.0
Midterm	30%	83 - 86	B Good	3.0
Final Exam	35%	80 - 82	B-	2.7
Total	100%	77 - 79	C+	2.3
*Assignment 01 from the Mid-term syllabus and Assignment 02 from the Final Term syllabus. ** In the count of Best 2, one from the mid-term and the other from the final term.		73 - 76	C Average	2.0
		70 - 72	C-	1.7
		67 - 69	D+	1.3
		60 - 66	D Poor	1.0

*Please Refer to the NSU Student Handbook, Section: "Grading Policy".

Course Schedule:

Lecture	Topics	Learning Activities	Assessment tools	Learning Outcome
1-2 (13.1)	Functions of two variables: drawing of the natural domain in R^2 -plane and 3D graph plotting	Lecture	Discussions Quiz-1	CO-1
3 (13.2)	Limits and Continuity	Lecture Assignment-01	Quiz-1	CO-1
4 (13.3)	Partial Derivatives: 1 st order derivatives and their physical significance	Lecture Group Discussion	Discussions Quiz-2 Mid-term	CO-1
5 (13.3)	Partial Derivatives: 2 nd and mixed order derivatives and their application	Lecture Discussion	Quiz-2	CO-1
6 (13.4&13.5)	Differentiability and Chain Rule	Lecture	Mid-term	CO-1
7 (13.6)	Directional Derivatives	Lecture Assignment-01	Mid-term	CO-1
8 (13.7)	Tangent planes and normal lines	Lecture	Mid-term	CO-1
9 (13.8)	Maxima and minima of functions of two variables	Discussion Lecture Assignment-01	Mid-term	CO-2
10 (14.1)	Double Integrals over rectangular regions	Lecture Assignment-01	Mid-term	CO-2
11-12 (14.2)	Double Integrals over non-rectangular regions	Lecture	Mid-term	CO-1
13	Mid-Term Exam			
14 (14.3)	Double Integrals in Polar Coordinates	Lecture	Quiz 3 Final Exam	CO-4
15 (14.5)	Triple Integrals in Cartesian Coordinates	Lecture Assignment-02	Quiz 3 Final Exam	CO-4
16-17 (14.7)	Change of variables in Multiple Integrals; Jacobian	Lecture Assignment-02	Quiz 4 Final Exam	CO-3
18 (11.8, 14.6)	Triple Integrals in Cylindrical and Spherical Coordinates	Lecture Assignment-02	Quiz 4 Final Exam	CO-3 CO-2
19 (15.1)	Vector fields	Lecture Assignment-02	Final Exam	CO-5

20 (15.2)	Line integrals: for scalar functions and vector fields	Lecture Assignment-02	Final Exam	CO-5
21 (15.4)	Green's Theorem	Lecture Assignment-02	Final Exam	CO-5
22 (15.5)	Surface Integrals: for scalar functions and vector fields	Lecture Assignment-02	Final exam	CO-3
23 (15.7)	Divergence theorem	Lecture Assignment-02	Final exam	CO-3
24 (15.8)	Stokes' theorem: (verification and evaluation of line integral)	Lecture, Presenting, Explaining, Demonstrating	Final exam Formative assessment	
Final Exam				

Rules and regulations:

- (a) There is **no scope to retake a quiz**. In case of a Mid-term or Final exam, exceptional cases (unfortunate physical inability, accidents, serious illness) may be considered conditionally (with a **penalty of 20% reduced marks**) with proper justification.
- (b) Three consecutive absences need an official clarification.
- (c) A student having attendance less than 60% of the total classes will not be allowed to sit for the **Final Exam**.

**** FOUR quizzes will be taken.**

***** Two assignments will be taken.**

******* No Make-Up Exam *******